

18.2 Group Control

1. Time Synchronization

- 1) On the master device's terminal, check the current system time by entering the command below:

```
date
```

```
> date  
Tue Oct 29 22:46:31 UTC 2024
```

- 2) On the slave device, synchronize its system time with the master's time by entering:

```
sudo date -s "2024-10-29 22:46:31"
```

```
> sudo date -s "2024-10-29 22:46:31"  
Tue Oct 29 22:46:31 UTC 2024
```

Note:

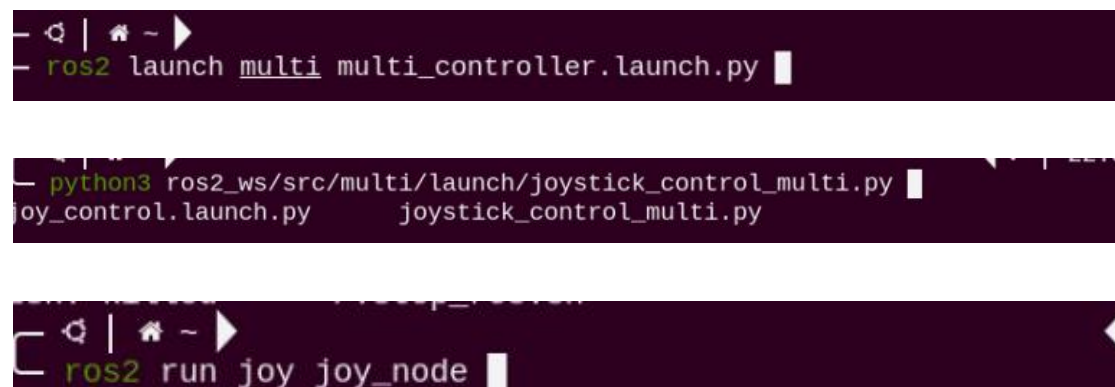
- ① If you experience noticeable desynchronization during joystick control between the master and slave devices, it is likely caused by inconsistent system times. In such cases, please repeat the synchronization process.
 - ② Updating the slave's system time based on the master's reported time takes some time to execute, meaning the master's actual time will have advanced by the time the update is applied. To minimize this discrepancy, start a stopwatch or manually count seconds after retrieving the master's time. Then, add this elapsed time to the master's reported time when setting the slave's time.
 - ③ After synchronizing, verify by quickly running the date command on both master and slave terminals. If their system times closely match, the synchronization is successful.
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2. Program Execution

1) On the master device, open three terminal windows and run the following programs in each:

- multi_controller.launch.py: controls the chassis
- joystick_control_multi.py: manages joystick input
- joy_node: starts the joystick communication node

```
ros2 launch multi multi_controller.launch.py
python3 ros2_ws/src/multi/launch/joystick_control_multi.py
ros2 run joy joy_node
```



2) On the slave device, enter the following command to run the program.

```
ros2 launch multi multi_controller.launch.py
```



Note: For every subsequent startup when you want to use multi-device control, you must first disable the auto-start services on both the master and slave devices, and synchronize their system times.

3. Group Control Operation

- 1) Turn ON the controller. Both the red and green LEDs on the controller will start blinking simultaneously.
- 2) Wait a few seconds for the robot and controller to pair automatically. Once paired successfully, the green LED will remain solid while the red LED turns off.
- 3) When the controller connects to the master device, pressing the **Start** button will trigger a beep from the buzzers on both vehicles.

Note: Gently pushing the joystick in any direction will result in slow-speed movement.

Key	Function	Operation
START	Stop and reset the robot	Press
Left joystick up	Move forward	Push
Left joystick down	Move backward	Push
Left joystick left	Turn left (mecanum wheel chassis only)	Push
Left joystick right	Turn right (mecanum wheel chassis only)	Push
Right joystick left	Turn left (Ackermann steering—front wheels only)	Push
Right joystick right	Turn right (Ackermann steering—front wheels only)	Push